

# OPTICALLY STIMULATED LUMINESCENCE OF EUROPIUM DOPED LAP

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## 1. Overview

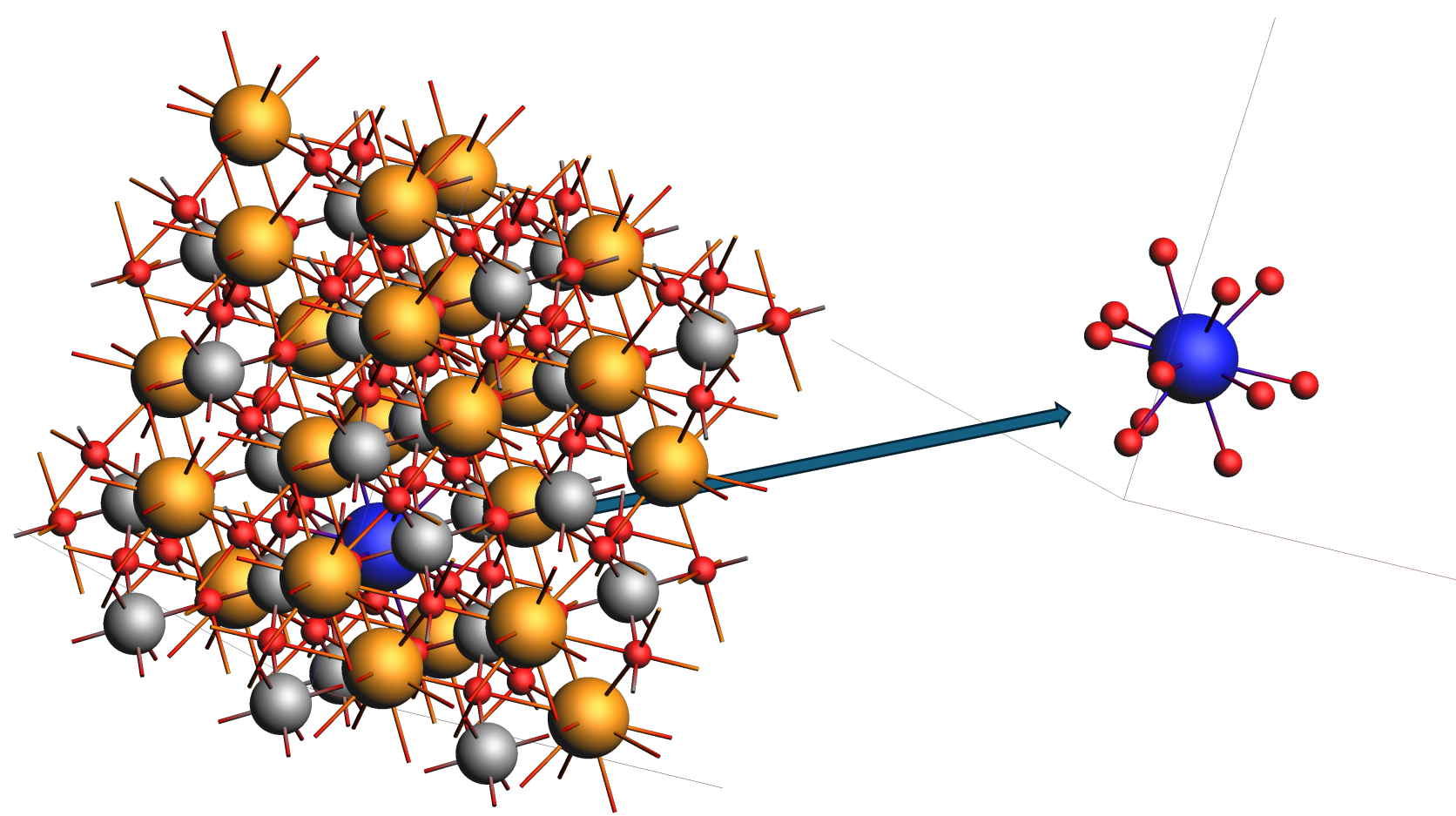
LaAlO<sub>3</sub>(LAP) doped with Europium is a phosphor widely used in lighting (LED) and display technologies and medical imaging, such as CT(computed tomography) and X-ray imaging.

In this work, we did the following:

- Compared the calculation of energy levels of Eu with the measured emission spectrum.
- Measured the decay dynamics of the excited Eu population after UV excitation.
- Evaluated the fit of a simple model.

## 2. Sample

### 1. Crstyal Structure of LaAlO<sub>3</sub>:Eu<sup>3+</sup>



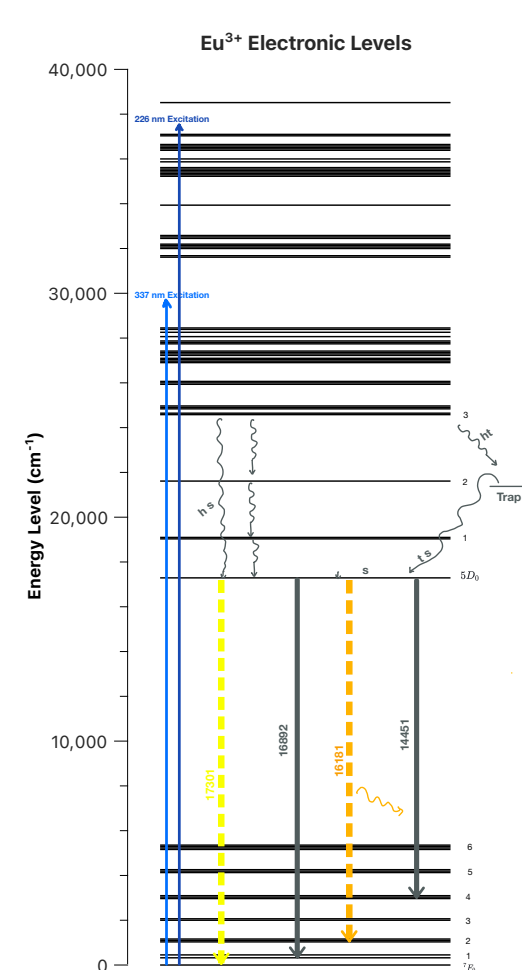
### 2. Mounted Sample



Room light

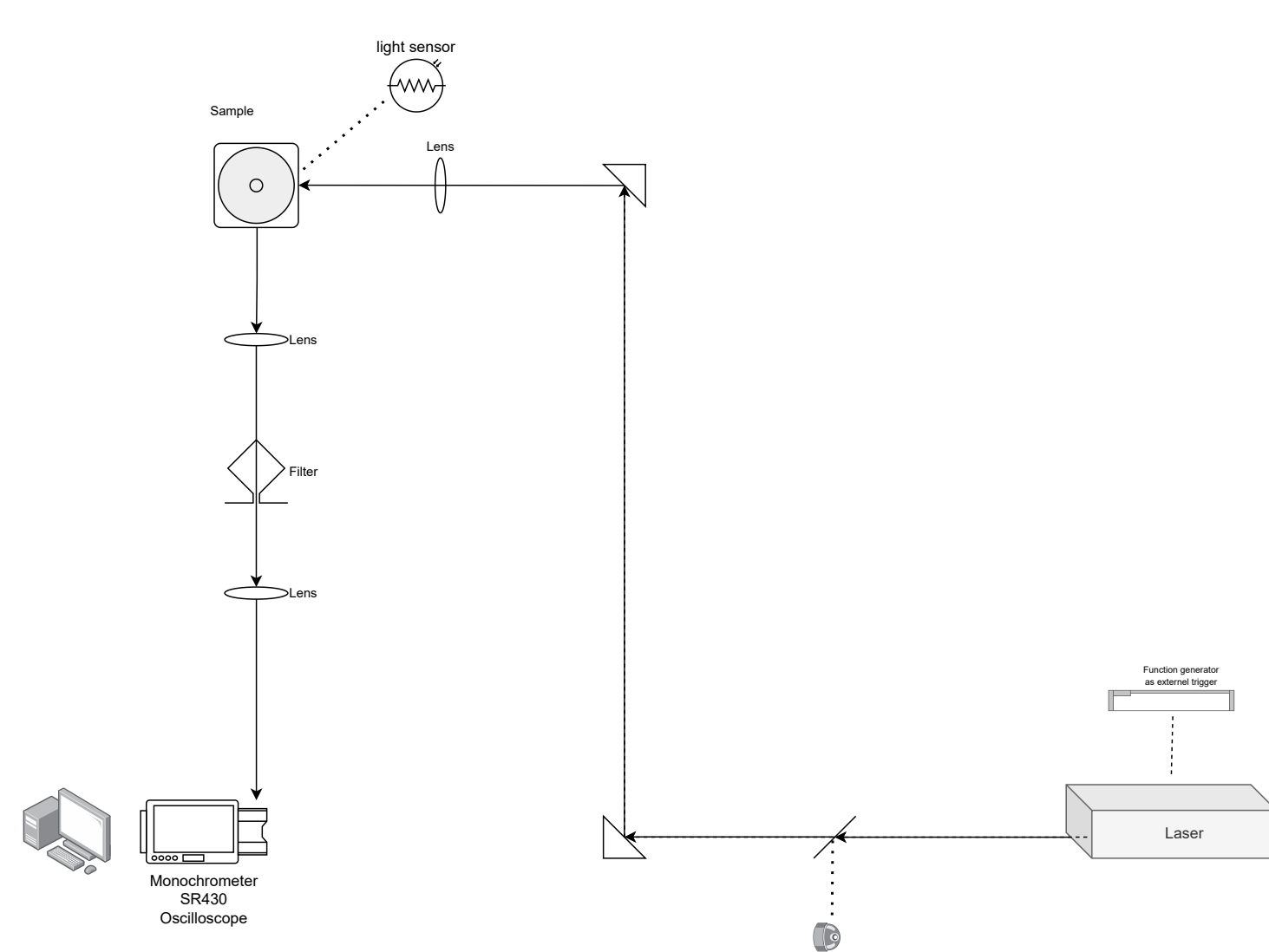
UV light

### 3. DFT Calculated 4f electronic levels



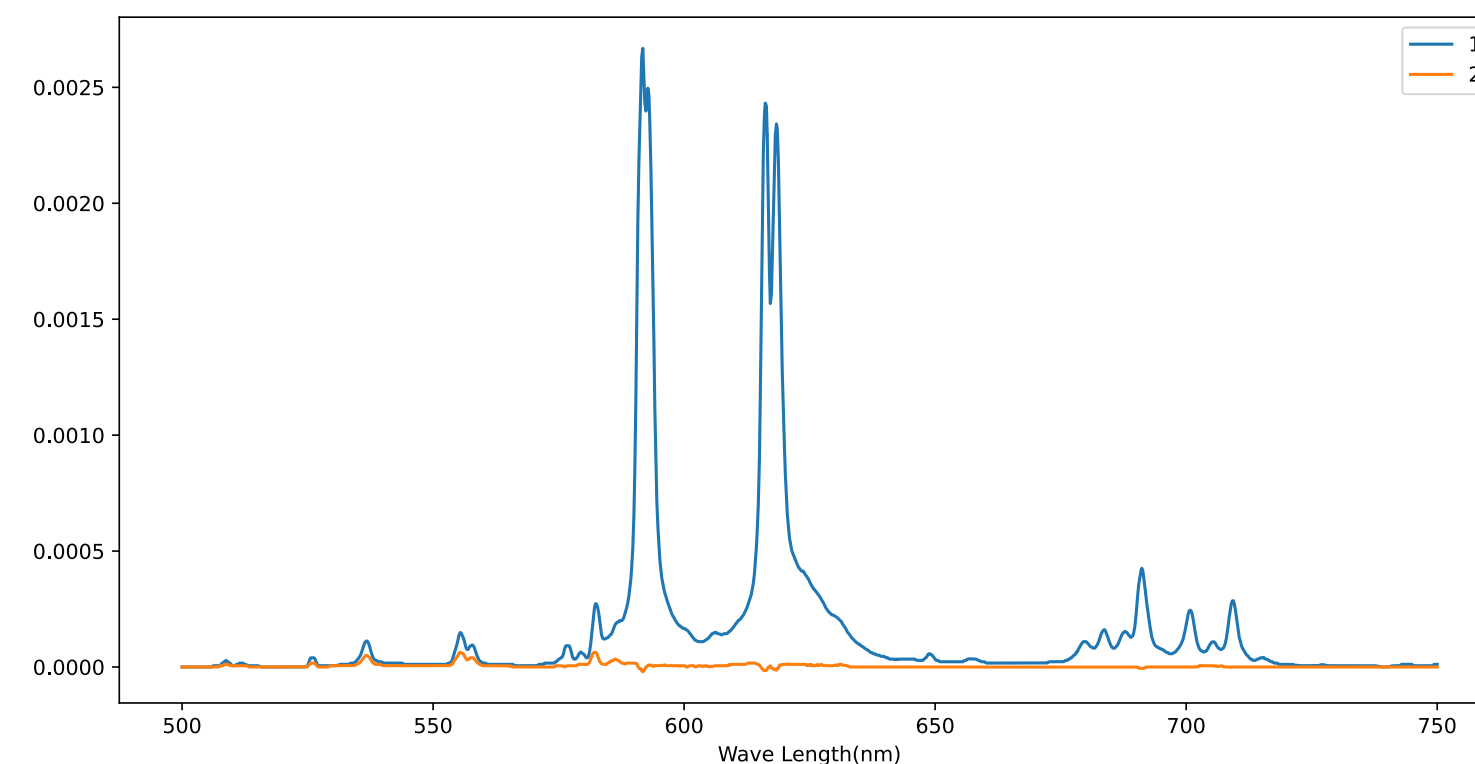
## 3. Experimental Setup

Measured the fluorescence of the Eu at 618 nm after pulsed excitation, with 337 nm and 266 nm.

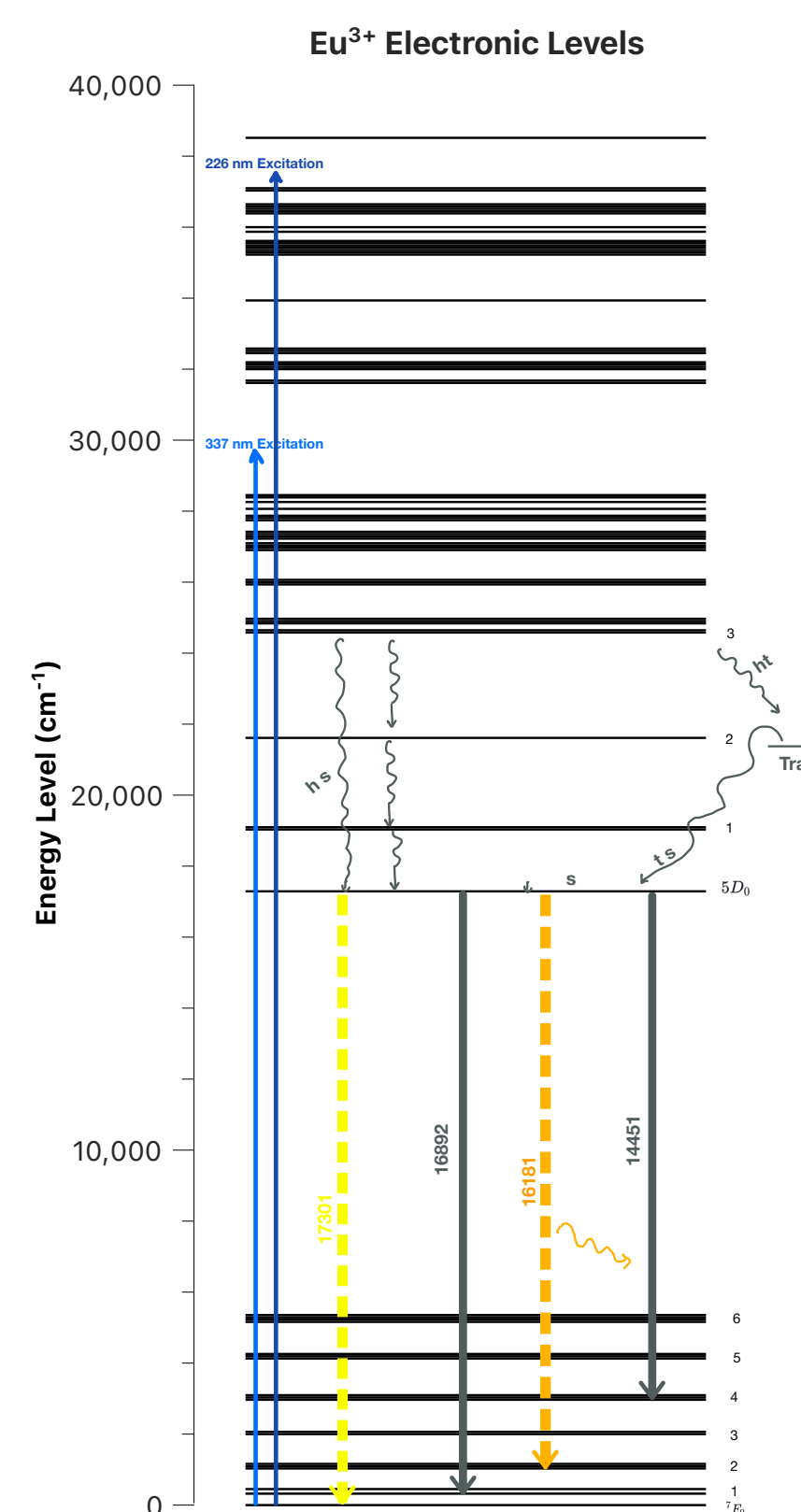


## 4. Measurement Results

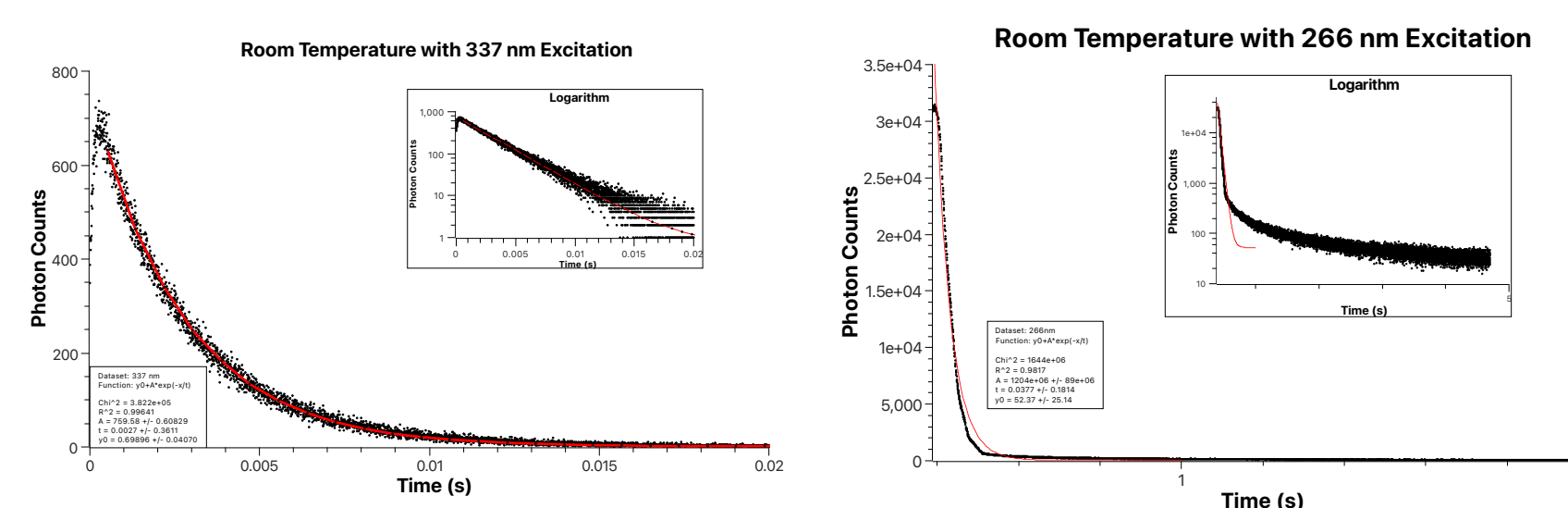
### 1. Emission (266 nm excitation)



### 2. Electronic Levels, transitions with charge trap model.



### 3. Typical lifetime measurement results

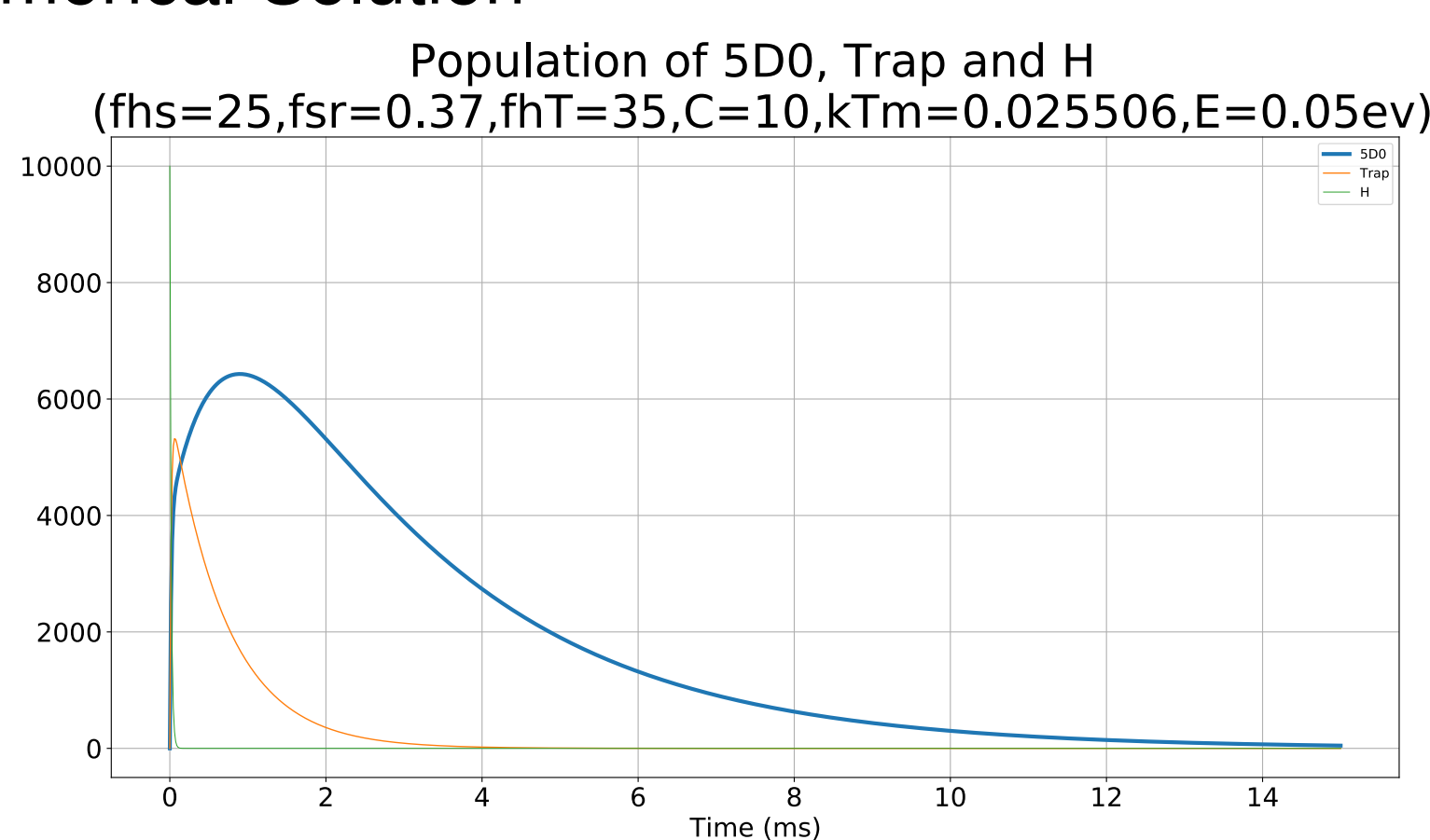


## 5. Proposed Model

$$\begin{aligned} \dot{S} &= Hf_{hs} + Tf_{TS} - Sf_{sr} \\ \dot{T} &= Hf_T - Tf_{TS} \\ \dot{H} &= -H(f_{hT} + f_{hs}) \end{aligned} \quad (1)$$

S is the population of Eu in the <sup>5</sup>D<sub>0</sub> excited state. T is the population of the traps H is the population of any higher energy state (Population: the number of Eu in each state)

### • Numerical Solution



Blue is the S and Orange is the Trap population

### • Analytical Solution

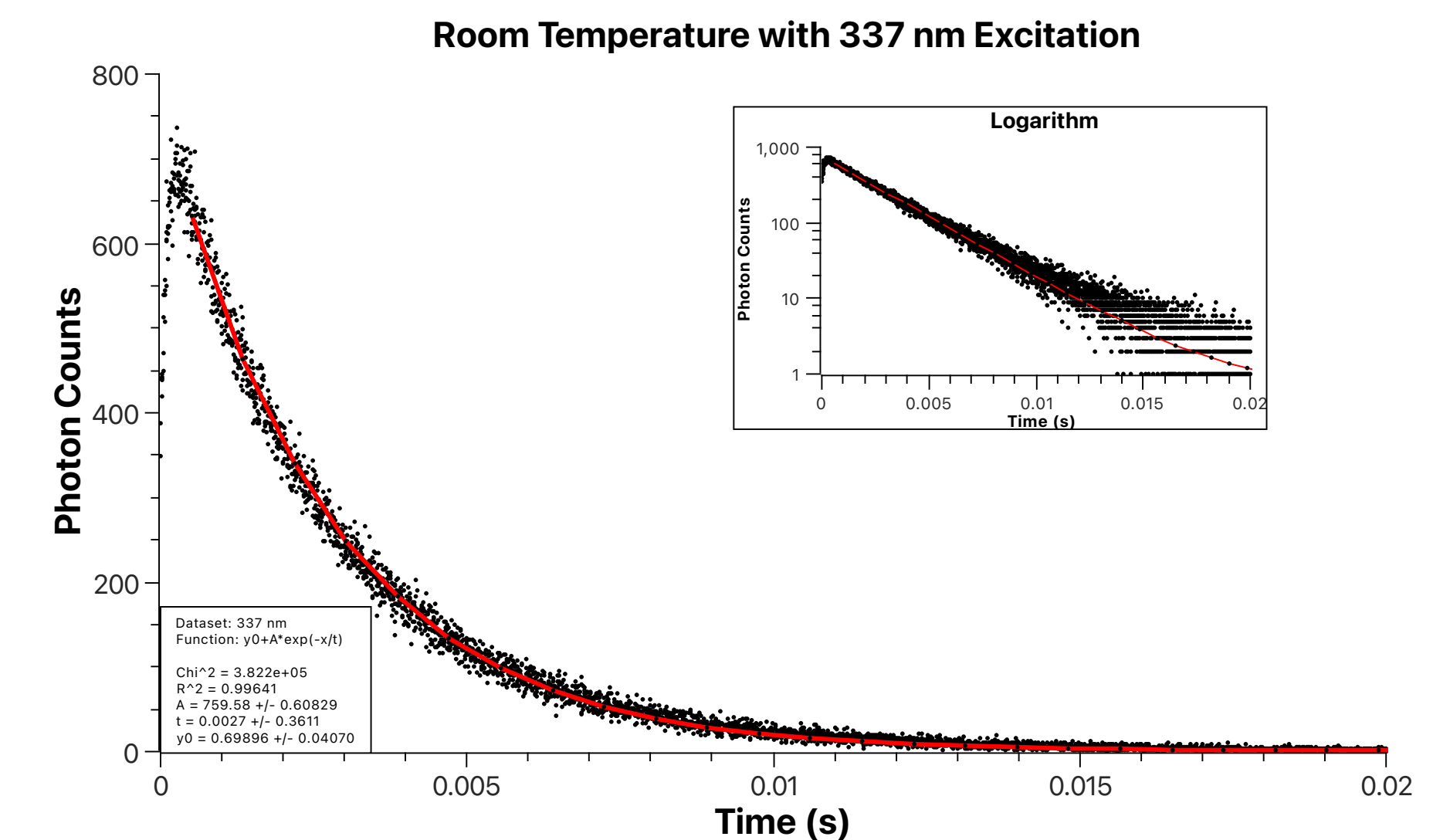
Ignore initial filling of S and T from H

$$S(t) = C_s f_{sr} e^{-f_{sr} t} + \left( \frac{f_{TS}}{f_{sr} - f_{TS}} \right) C_T e^{-f_{TS} t} \quad (2)$$

$$\frac{dS}{dt} = -k_1 e^{-f_{sr} t} - k_2 e^{-f_{TS} t} \quad (3)$$

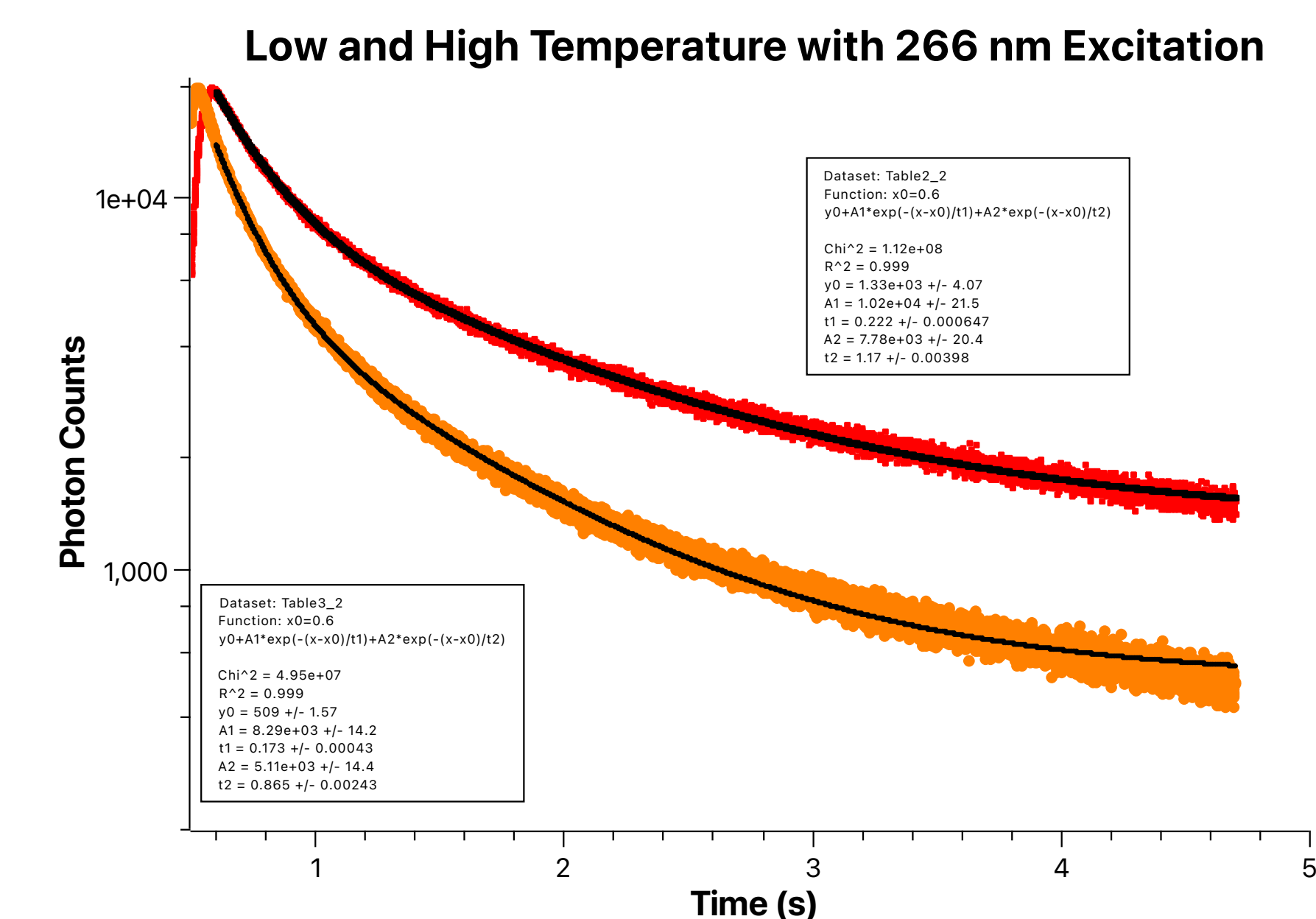
## 6. Model Fit with 337 nm

Asuming  $k_2 \ll k_1$  and  $f_{TS} \gg 1$



- The 337 nm excitation does not contribute to phosphorescence.
- Radiative lifetime of Europium is 2.7 ms.
- The rise of population after pulse is observed.

## 7. Model Fit with 226 nm, 78K, 250K...



- The biexponential model fits well.
- The phosphorescence strongly depends on temperature, with 35% increase in the T to S transfer rate ( $f_{TS}$ ) over the 78-255K range.

## 8. Conclusions

- Sample of LaAlO<sub>3</sub>:Eu<sup>3+</sup> have been made by combustion synthesis.
- The emission spectrum was used to identify the transition between 4f levels and compared to DFT calculation with good agreement
- The decay of both phosphorescence and fluorescence emission has been measured.
- Both emissions are well-described by simple biexponential model.
- The trap to impurity transfer rate has been found to decrease at lower temperatures.

## References

- [1] M. Lastusaari P. Gluchowski, W. Strek and J. Holsa. Optically stimulated persistent luminescence of europium-doped LaAlO<sub>3</sub> nanocrystals. *Phys.Chem.Chem.Phys.*, 17:17246–17252, 2015.
- [2] C. Furetta and G. Kitis. Review Models in thermoluminescence. *Journal of Materials Science*, 39:2277–2294, 2004.



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